`quap()` – aproximação quadrática

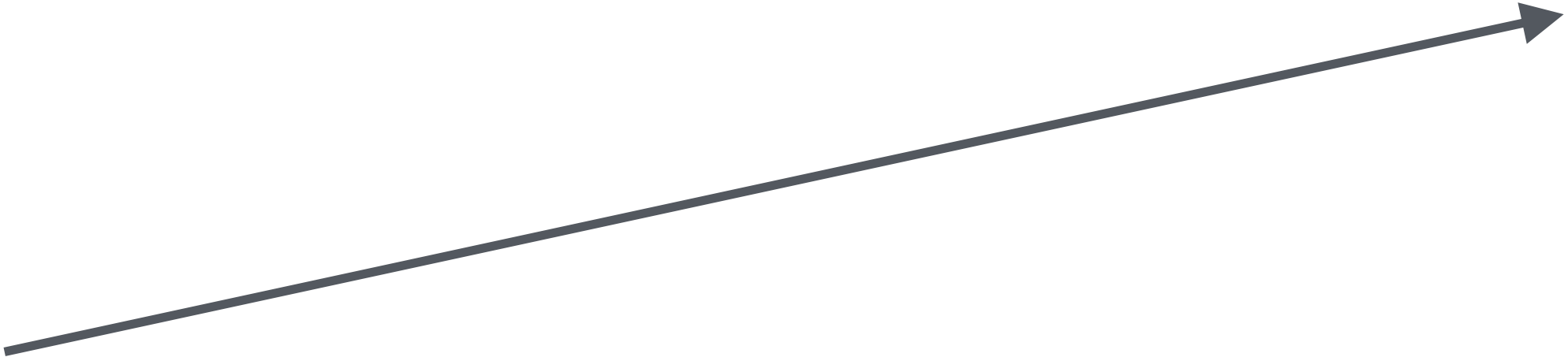
Amostradores no Rethinking

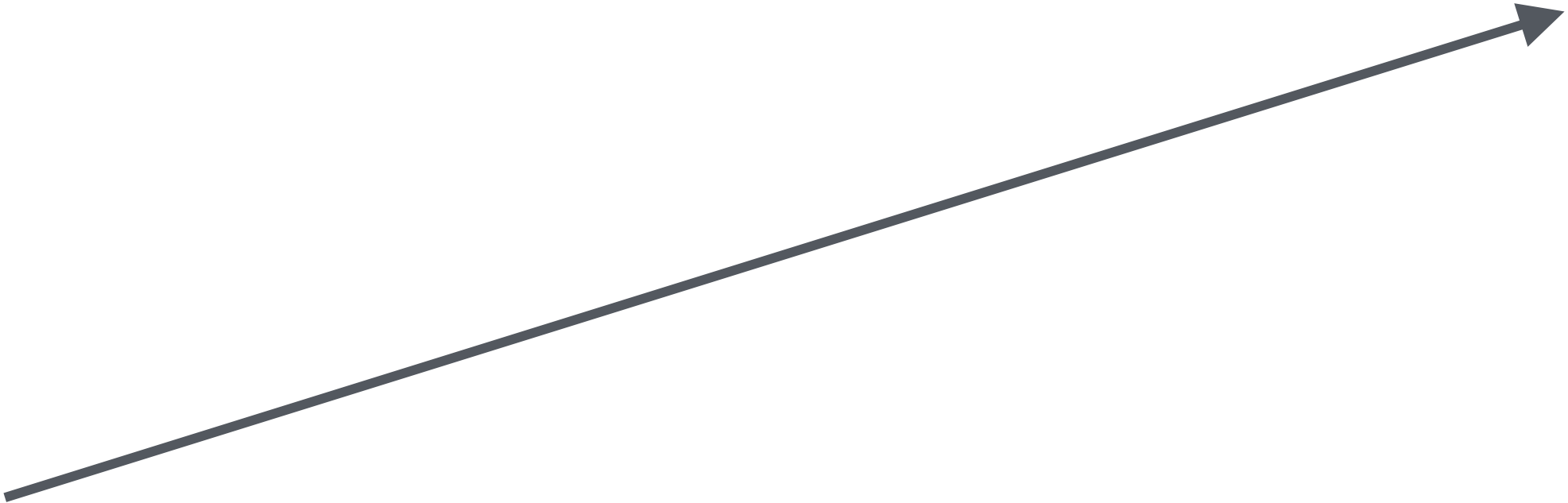
```r
# Data
library(rethinking)
d2 <- Howell1[ Howell1$age >= 18 , ]

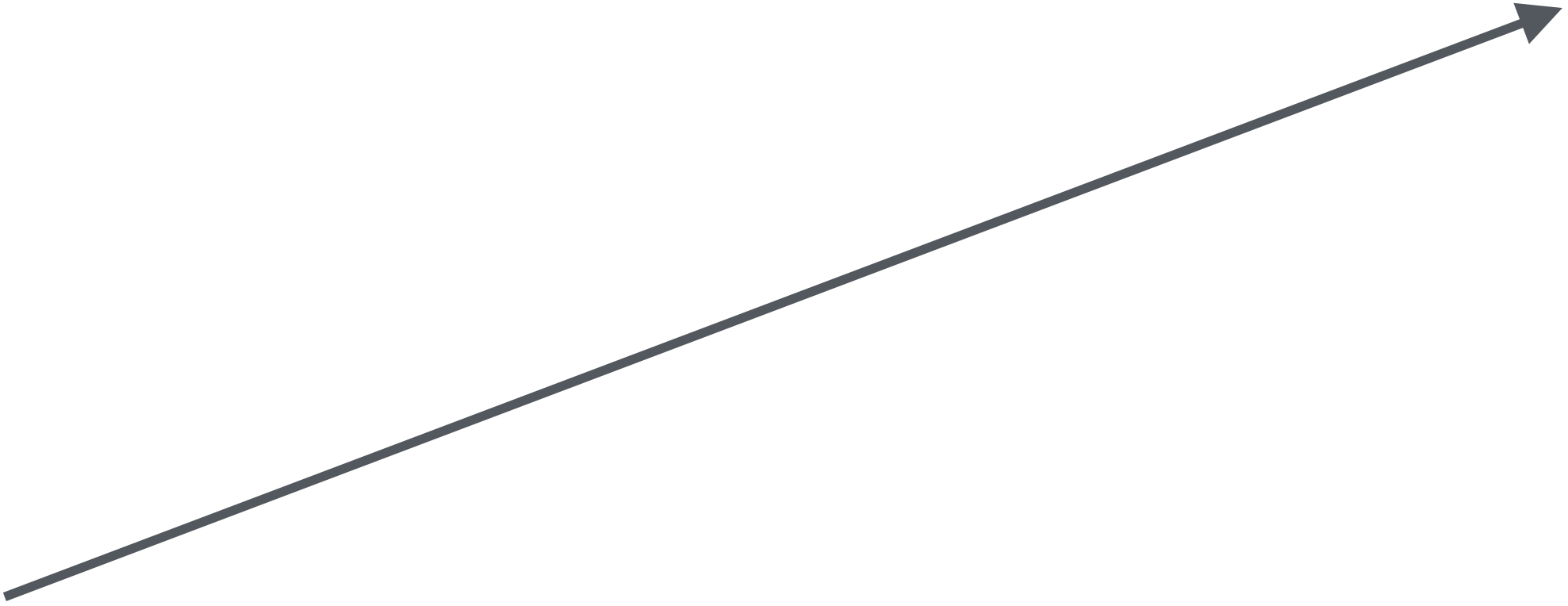
# Model
quap(alist(
  y ~ dnorm(mu, sigma),
  mu <- a + b * x,
  a ~ dnorm(0, 20),
  b ~ dlnorm(0, 1),
  sigma ~ dexp(1)),
  data = list(y = d2$height,
              x = d2$weight))
```











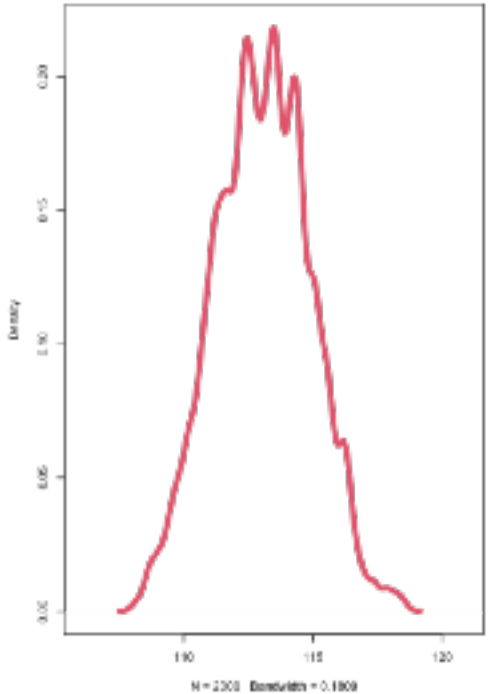
$$y_i \sim Normal(\mu_i, \sigma)$$

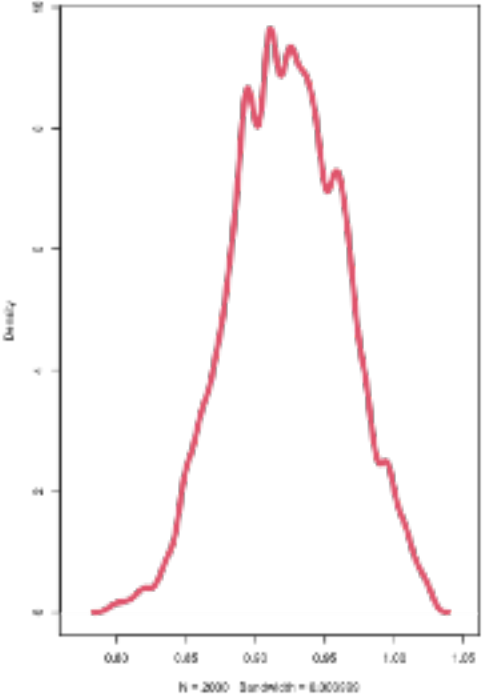
$$\mu_i = \alpha + \beta x_i$$

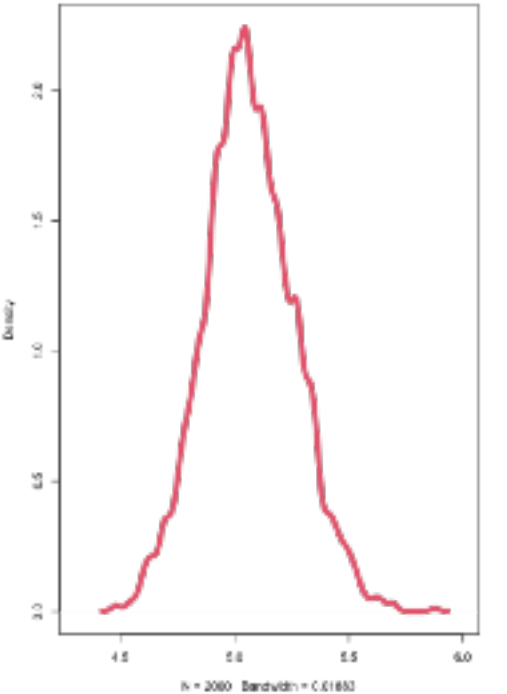
$$\alpha \sim Normal(0, 20)$$

$$\beta \sim lognormal(0, 1)$$

$$\sigma \sim Exponential(1)$$

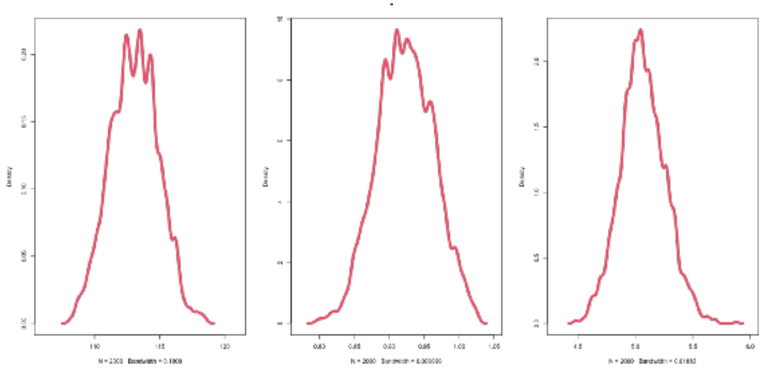

Density
110
115
120
N = 2000 Bandwidth = 0.1809


Density
0.80
0.85
0.90
0.95
1.00
1.05
N = 2000 Bandwidth = 0.003939


Density
4.5
5.0
5.5
6.0
N = 2000 Bandwidth = 0.01853

Amostradores no Rethinking

`quap()` – aproximação quadrática



$$y_i \sim Normal(\mu_i, \sigma)$$
$$\mu_i = \alpha + \beta x_i$$
$$\alpha \sim Normal(0, 20)$$
$$\beta \sim lognormal(0, 1)$$
$$\sigma \sim Exponential(1)$$

```
# Data
library(rethinking)
d2 <- Howell1[ Howell1$age >= 18 , ]

# Model
quap(alist(
  y ~ dnorm(mu, sigma),
  mu <- a + b * x,
  a ~ dnorm(0, 20),
  b ~ dlnorm(0, 1),
  sigma ~ dexp(1)),
  data = list(y = d2$height,
              x = d2$weight))
```

Amostradores no Rethinking

`ulam()` – Hamiltonian Monte Carlo

$$
\begin{aligned}
y_i &\sim Normal(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta x_i \\
\alpha &\sim Normal(0, 20) \\
\beta &\sim lognormal(0, 1) \\
\sigma &\sim Exponential(1)
\end{aligned}
$$

```
# Data
library(rethinking)
d2 <- Howell1[ Howell1$age >= 18 , ]

# Model
ulam(alist(
  y ~ normal(mu, sigma),
  mu <- a + b * x,
  a ~ normal(0, 20),
  b ~ lognormal(0, 1),
  sigma ~ exponential(1)),
  data = list(y = d2$height,
              x = d2$weight),
  iter = 1000, chains = 4, cores = 4)
```